IIT Mandi

Office of the Dean of Resources & Alumni Affairs (DORA)

Project Proposal and Working Agreement

Automated Ranking Portal

(ARP)

An Institutional Ranking Data Aggregation, Processing, and Analytics Platform

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A proposal submitted for consideration and approval by the
Office of the Dean of Resources & Alumni Affairs (DORA), IIT Mandi

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Submitted by:

Student Development Team
(Team of 4 Students)

1. _____
2. _____
3. _____
4. _____

Submitted to:

Office of the Dean of Resources & Alumni Affairs (DORA)
IIT Mandi

Date: September 21, 2026

This document constitutes a proposal and, upon signature by both parties, a working agreement governing scope of work, timelines, reporting, and compensation for the project described herein.

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1 Introduction and Project Overview

The data collection and compilation process for institutional ranking submissions (such as NIRF, QS World University Rankings, and Times Higher Education) at IIT Mandi is currently a manual pipeline operated by administrative staff. This process involves gathering student, faculty, and learnability data from various departments, manually identifying errors, and re-formatting this data into agency-specific formats.

Because data arrives via email in multimodal formats (Excel sheets, Word documents, and PDFs), the process is highly time-consuming and error-prone. This document proposes the design and iterative development of a web-based platform, the **Automated Ranking Portal** ("ARP"/"the Platform"), by a team of four (4) students ("the Development Team"/"the Team") under the guidance of DORA.

The Platform covers three connected areas of DORA's work:

1. **Data Collection and Governance**: Department nodal officer management, multimodal document ingestion via the portal, secure document repositories for proofs, and automated deadline tracking.

2. **Processing and Validation**: Automated data unification, AI-assisted error spotting, maker-checker correction workflows, and audit trails for all data modifications.

3. **Export and Analytics**: Agency-specific template generation (NIRF, QS, THE), executive dashboards for historical tracking, and department-wise contribution/performance analytics.

While the idea is simple to state, the underlying logic involves substantial data schema unification and pipeline complexity (Section 2), so the Platform is best understood as an extensive data governance and compliance engine coupled with a leadership-facing analytics web application. This document also serves as a working agreement defining scope, reporting, review, and compensation for the Development Team.

2 Scope of Work

The scope below reflects the current high-level understanding of requirements. As described in Section 3.1, this scope is expected to evolve as DORA raises new requirements during the engagement.

Part A — Department Data Collection and Governance

2.1 Institute Organogram and Hierarchy Mapping

- Creation of an interactive institute organogram mapping academic, administrative, and functional sections/subsections.

- Structural representation ensuring data collection is managed according to the exact hierarchical organization of the institute.

2.2 Targeted Admin-Based Data Collection

- Admin interface to manage the roster of designated Nodal Officers for each department and section.

- Capability for the admin to select specific sections based on the institute organogram, identify required data, and dispatch targeted email requests.

- Automated provisioning of restricted portal access for Nodal Officers to securely upload required ranking proofs.

2.3 Flexible Data Ingestion Engine

- Secure submission of all department data exclusively through the portal by authenticated Nodal Officers, eliminating dependency on email-based collection.
- Submission of required data through Platform-specific web forms and standardised Excel templates, with supporting documents uploaded alongside as evidence.
- Conversion of multimodal raw data into a single, unified internal schema that maps institutional metrics across different ranking methodologies.

2.4 Secure Document Proof Repository

- A centralized, searchable repository to store all raw documents submitted by departments (e.g., scanned proof of research grants, faculty awards) ensuring permanent evidence linkage.
- Strict data provenance mapping: every single data point must be permanently linked to its source proof (e.g., clicking on a "Sponsored Research Revenue" figure allows the auditor to instantly trace its origin and view the exact grant sanction PDF).
- Document versioning to ensure historical records are maintained when a department submits updated or corrected files.

2.5 Deadline and Compliance Tracking

- Admin-defined submission cycles with specified timelines and hard deadlines for data submissions.
- System tracking of exactly which section has been asked for data, what specific data was requested, expected deadlines, and receipt status.
- Automated and manual email alerts/reminders dispatched to Nodal Officers whose submissions are pending or incomplete.
- Compliance dashboard for DORA to view, at a glance, the overall progress of the ranking submission process.

Part B — Data Processing, Unification and Validation

2.6 Historical Data Querying and Value Comparison

- Storage and easy backend querying of at least the previous three years of historical ranking data for institutional reference.
- Real-time contextual display of historically submitted values alongside new data entries during updates (e.g., Previous: X, New: Y, Change: Increase/Decrease).
- Immediate visual feedback to the data entrant and reviewer, simplifying data verification before submission.

2.7 Faculty and Student Roster Normalization

- Standardization of faculty data across departments (e.g., normalizing variations in designation, degree qualifications, and publication indexing details).

- Aggregation of student demographics, intake numbers, graduation rates, and placement statistics into master institutional metrics.
- Automated deduplication logic to prevent double-counting of interdisciplinary faculty or joint research projects.

2.8 Automated Error Spotting and Rule-Based Validation

- Deterministic rule-based checks applied instantly upon data ingestion (e.g., ensuring 'Total PhDs Awarded' cannot be negative, or 'Placement Percentage' cannot exceed 100).
- Automatic flagging of missing mandatory columns, empty rows, or corrupted data cells in uploaded Excel sheets.
- Visual highlighting of errors within the admin dashboard, allowing the reviewer to quickly trace the error back to the exact source document.

2.9 AI-Assisted Discrepancy Flagging

- Heuristic and AI-driven analysis comparing current year submissions against historical trends to detect statistical anomalies.
- Example: Automatically flagging a department if their reported research publications drop by 80% compared to the previous year, prompting administrative review.
- Severity grading of flags (e.g., Critical vs. Warning) to help administrators prioritize data auditing efforts.
- **Reliability Principle and Grounding:** The AI layer is an enhancement, never a dependency. The system is designed to work safely with sensible fallbacks if AI services are unavailable. All AI flags must be grounded in approved portal data, avoiding hallucinated policies or numbers.

2.10 Maker-Checker Workflow for Data Corrections

- When an error is flagged, the Platform allows DORA to formally raise a "Correction Request" directed back to the department's Nodal Officer.
- The Nodal Officer (Maker) submits the corrected data or overriding justification via the portal.
- DORA (Checker) reviews the updated submission and explicitly clicks "Approve" or "Reject", recording the decision. Data is not finalized until Checker approval.

2.11 Audit Log and Data Change Tracking

- Immutable recording of every create, edit, approve, reject, and export action across the platform, including the old value, new value, actor, and timestamp, guaranteeing complete data provenance.
- Dedicated interface for internal auditors to filter, search, and export the audit log to ensure data integrity prior to external submission.

Part C — Agency Export and Ranking Calculation

2.12 Agency-Specific Schema Mapping

- Maintenance of internal configuration maps that translate the unified institutional data into the exact nomenclatures and structures demanded by specific agencies.
- Support for dynamic parameter adjustments, as ranking agencies frequently update their required data formats year-over-year.

2.13 Final Data Packet Generation and Export

- Automated compilation of the approved data into the required final formats (e.g., NIRF Data Capturing System (DCS) Excel templates).
- Generation of consolidated PDF summary reports for the Director's final sign-off before upload.
- Supported export targets include **NIRF** (all 4 categories prioritized), **MDRA** (Best University & Engineering), **QS World University Rankings**, and **Times Higher Education (THE)**.

Part D — Insights, Dashboards and Analytics

2.14 Public-Facing Ranking Data & Visualizations

- A publicly accessible portal requiring no administrative or developer access, allowing any visitor to view institutional ranking data.
- Automated fetching and visualization of year-wise ranking trends across different ranking systems to show performance over time.
- Visual comparative analytics benchmarking IIT Mandi's performance against other relevant institutions, particularly 2nd-generation IITs.

2.15 Executive Ranking Dashboards

- A high-level, leadership-facing dashboard visualizing IIT Mandi's historical ranking performance over a selectable time range (e.g., last 5 years).
- Parameter-wise breakdown showing exactly where the institute gained or lost points (e.g., Teaching & Learning vs. Research & Professional Practice).
- Visual gap analysis highlighting metrics where the institute is falling behind peer benchmarks or its own historical averages.

2.16 Department-wise Contribution/Performance Analytics

- Granular dashboards that analyze internal department contributions to the overall institutional score (e.g., which department brought in the most research funding or had the highest placement rate).
- Ability for department heads to view their own analytics to identify areas of excellence and areas requiring improvement.

2.17 Automated Briefing Generation

- Ability to generate a standardized, boardroom-ready PDF briefing detailing the year's ranking submission statistics, major discrepancies caught by the system, and projected ranking outcomes (which are scenario-based estimates only).

Part E — Cross-Cutting Requirements

2.18 Roles and Access Control

The Platform will support strict role-based access control (RBAC) as follows:

- **Admin/Staff (Data Officer):** Full access to trigger ingestion, review and approve Nodal Officer submissions, initiate correction requests, and generate agency exports.
- **Department Nodal Officer:** Restricted access to upload data/proofs for their specific department only, and to respond to assigned correction requests. Cannot view other departments' raw data.
- **Leadership/Director:** Read-only access to all executive dashboards, analytics, and final generated reports.
- **Auditor:** Read-only access to raw proofs, the audit log, and historical submission trails to ensure compliance.

2.19 Data and System Inputs Required from DORA

To build and operate the Platform, the Development Team will require the following from DORA:

- **Historical ranking data** and previous agency-specific schema formats for NIRF, QS, and THE.
- **Sample raw data** (Excel, Word, PDF) typically received from departments to train and test the multimodal ingestion pipeline.
- **Authorised email account access:** access to, or a delegated permission on, the authorised email account receiving department submissions.
- **Infrastructure support:** guidance on hosting, server, and cloud backup requirements. Where specialized support is needed, DORA will direct the Team to the concerned person/department (e.g., IT Services).

Confidentiality of institutional data. All data shared by DORA for the purposes of building and operating the Platform is proprietary and confidential to the Institution. The Development Team shall not share, disclose, publish, or use this data outside the scope of this engagement, and shall not share it with any third party, under any circumstance, without the Institution's prior written consent.

2.20 Out of Scope (Initial Phase)

- Direct, automated API integration to upload data *into* the NIRF/QS/THE portals, as these agencies typically require manual authenticated uploads by the Director.
- Historical data entry or digitization of unstructured paper records from previous decades.

Any new feature or change requested by DORA beyond the scope above will be evaluated for feasibility. If accepted for inclusion, any change in scope, cost, or timeline shall require written prior approval from DORA before implementation begins.

3 Development Approach

3.1 Iterative Development

A complete, final system architecture cannot be reasonably fixed at the outset given the complexity in Section 2. The Platform will therefore be developed iteratively: new requirements raised by DORA will be assessed for feasibility (Section 2.20). Any resulting changes to the working scope, project timeline, or compensation shall strictly require prior written approval and will be documented as a signed addendum to this agreement.

3.2 Version Control and Work Tracking

All source code will be maintained in an Institute-controlled (DORA-owned) GitHub repository from the outset. All work will be tracked and submitted via GitHub Pull Requests (PRs), which serve as the primary record of individual contribution. The platform and its data will be hosted exclusively on Institute-provisioned servers.

3.3 Institutional Deployment and Production Readiness

In line with best practices for institutional systems, the Platform will pass through stringent production gates before real data is loaded:

- **Ownership and Control:** IIT Mandi retains full control of the production database, storage, domains, deployment accounts, and the underlying data schema from day one.
- **Reproducible Handover:** The team will deliver installation, configuration, backup/restore, and deployment runbooks, ensuring an Institute technical owner can independently manage staging and production instances.
- **Controlled Pilot and Go/No-Go:** Before a campus-wide launch, the system will undergo a controlled pilot with selected departments to validate KPIs, followed by a formal go/no-go decision. The pilot must strictly meet the following **Measurable KPIs**:
  - *AI Reliability:* >95% accuracy in flagging data discrepancies, with 0% false positives on critical institutional metrics.
  - *Data Auditability:* 100% of data modifications must be accurately tied to a recorded user session and timestamp in the immutable audit log.
  - *System Resilience:* A witnessed disaster recovery drill must confirm the target 24-hour RPO and 8-hour RTO are achievable.

4 Team Structure

The Development Team shall consist of four (4) students, listed below (to be finalized and appended to this document upon submission):

1. _____ (Team Lead / Point of Contact)
2. _____ (Team Lead / Point of Contact)
3. _____
4. _____

Team members 1 and 2 shall jointly serve as Team Leads and primary points of contact with DORA for scheduling meetings, consolidating weekly reports, and coordinating delivery.

5 Timeline and Prioritized Milestones

Based on administrative requirements, the development pipeline is strictly prioritized to deliver maximum immediate value. The timeline is divided into four distinct phases, targeting an overall completion window ending **7 December 2026** (Section 10).

5.1 Phase 1: Core NIRF Workflow (High Priority)

- **Deliverable:** A functional end-to-end system for the NIRF data collection, validation, approval, and export lifecycle.
- **Scope:** Institute organogram setup, admin targeted data requests, node-officer upload portal, maker-checker validation workflow, and automated generation of the NIRF DCS Excel exports.
- **Acceptance Criteria:** The system successfully generates a complete, compliant NIRF export based on mock data inputs, and the maker-checker audit trail correctly logs all data modifications.
- **Estimated Effort:** 80 person-hours per week (20 hours per member).

5.2 Phase 2: Historical Analytics & Multi-Ranking Export

- **Deliverable:** Executive analytics dashboards and integration for additional ranking agencies.
- **Scope:** QS, THE, and MDRA schema mapping and exports. Department-wise contribution/performance analytics, and automated boardroom-briefing generation.
- **Acceptance Criteria:** Leadership dashboards accurately visualize historical trends and gap analyses. The platform can export valid templates for at least one additional agency (e.g., QS).
- **Estimated Effort:** 80 person-hours per week (20 hours per member).

5.3 Phase 3: Public Visualizations & System Hardening

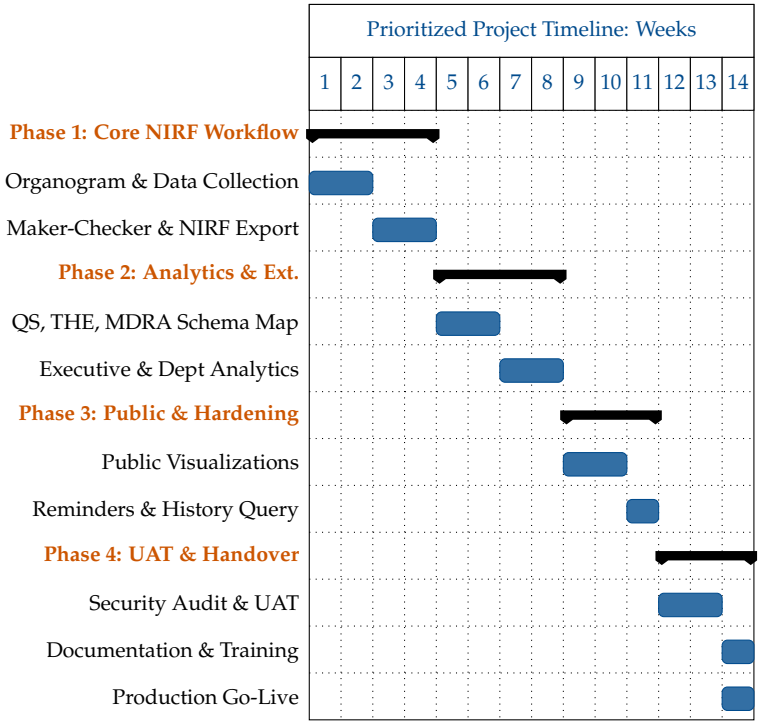
- **Deliverable:** Public-facing visualization portal and advanced system compliance.
- **Scope:** Public ranking trend visualizations, competitor benchmarking. Full implementation of automated reminders, contextual historical-value comparisons during data entry, and comprehensive RBAC testing.
- **Acceptance Criteria:** Public portal is accessible without authentication and accurately plots dummy historical data. Automated reminders trigger appropriately based on predefined deadlines.
- **Estimated Effort:** 60 person-hours per week (15 hours per member).

5.4 Phase 4: UAT, Handover, and Post-Deployment Support

- **Deliverable:** Production-ready deployment, complete documentation, and trained staff.
- **Scope:** Detailed User Acceptance Testing (UAT) with actual DORA staff, comprehensive technical and user manual documentation, training sessions for nodal officers, and a defined post-deployment support period. **Addition of Controlled Pilot:** A closed pilot with a selected subset of departments, allowing KPI measurement before a full migration and

go/no-go decision.

- **Acceptance Criteria:** DORA staff formally signs off on UAT. Handover of the Institute-controlled repository is complete. User manuals are provided. The service is functionally operable without dependency on the Development Team.
- **Estimated Effort:** 40 person-hours per week (10 hours per member).



Note: this Gantt chart illustrates the sequence of the prioritized phases. Actual phase durations, dependencies, and start dates will be agreed with DORA and revised as scope evolves under Section 2.20. Please see Appendix A for a detailed Milestone Matrix.

6 Reporting, Review, and Communication

6.1 Weekly Progress Reports

The Development Team shall submit one (1) written progress report per week, summarizing work completed, work in progress, and blockers, attributed by individual member and referencing the corresponding GitHub Pull Requests (Section 3.2).

6.2 Weekly Demonstration Meeting

The Development Team and a designated Administrative representative shall meet at most once per week to demonstrate progress and gather inputs. A meeting is not mandatory every single week; it will be scheduled as needed, by mutual convenience, particularly if there is no material update since the previous meeting.

6.3 Administrative Review and Delegation

Final data exports are generated only after review and approval by DORA.

6.4 Documentation of Communication

All key communication and decisions exchanged between the Team and DORA, including outcomes of meetings, will additionally be documented over email, serving as minutes of meeting (MoM) for record-keeping.

7 Compensation and Payment Terms

7.1 Stipend

Each member of the Development Team shall be compensated at a rate of Rs. 270 per hour of documented, verifiable work. To ensure budget predictability, the following weekly compensation ceiling is established (based on a maximum of 20 billable hours per member):

Resource	Max Hours / Week	Max Budget / Week
Per Team Member	20 Hours	Rs. 5,400
Total Team (4 Members)	**80 Hours**	**Rs. 21,600**

Any hours exceeding this ceiling must receive prior written approval from DORA.

7.2 Estimated Cumulative Cost

Based on the phase-wise estimated efforts and the 14-week project timeline, the projected cumulative cost of the Development Team is calculated comprehensively as follows:

Phase	Duration	Calculation (Members $\times$ Hrs/Wk $\times$ Wks)	Total Hrs	Est. Cost
Phase 1	4 Weeks	4 Members $\times$ 20 Hrs/Wk $\times$ 4 Wks	320 Hours	Rs. 86,400
Phase 2	4 Weeks	4 Members $\times$ 20 Hrs/Wk $\times$ 4 Wks	320 Hours	Rs. 86,400
Phase 3	3 Weeks	4 Members $\times$ 15 Hrs/Wk $\times$ 3 Wks	180 Hours	Rs. 48,600
Phase 4	3 Weeks	4 Members $\times$ 10 Hrs/Wk $\times$ 3 Wks	120 Hours	Rs. 32,400
Total	**14 Wks**	**Calculated at a rate of Rs. 270/hr**	**940 Hrs**	**Rs. 253,800**

Detailed Billable Hours Breakdown:

- **Phase 1 (Weeks 1–4):** Core NIRF Workflow requires maximum intensity. 4 members working up to the 20 hours/week ceiling over 4 weeks equates to $4 \times 20 \times 4 = 320$ hours. At Rs. 270/hr, the cost is Rs. 86,400.
- **Phase 2 (Weeks 5–8):** Analytics & Extensions continue at peak effort. 4 members $\times$ 20 hours/week $\times$ 4 weeks equates to $4 \times 20 \times 4 = 320$ hours. Cost is Rs. 86,400.
- **Phase 3 (Weeks 9–11):** Hardening phase operates with slightly reduced intensity. 4 members $\times$ 15 hours/week $\times$ 3 weeks equates to $4 \times 15 \times 3 = 180$ hours. Cost is Rs. 48,600.
- **Phase 4 (Weeks 12–14):** UAT and Handover require only maintenance-level engagement. 4 members $\times$ 10 hours/week $\times$ 3 weeks equates to $4 \times 10 \times 3 = 120$ hours. Cost is Rs. 32,400.

Note: This cumulative cost is an estimate based on projected efforts. Payment is strictly based on actual verifiable hours worked, capped at the absolute maximum project ceiling of 1,120 hours (Rs. 3,02,400) for the 14-week period.

7.3 Non-Monetary Compensation

In addition to the hourly stipend, any development tool or software subscription (e.g., Claude Code access) shall be procured or reimbursed only with prior written approval and strictly as per Institute rules.

7.4 Payment Schedule

Compensation shall be calculated and disbursed on a monthly basis. Payment for hours worked in a given calendar month shall be released during the first week of the following month, subject to submission and administrative acceptance of that month's weekly reports.

7.5 Integrity of Hours and Basis for Payment

Payment in a given month is contingent on submission of that month's weekly reports and verifiable work reflected through GitHub activity (Section 3.2). The Development Team commits to logging hours with utmost sincerity and honesty; no inflated, false, or unworked hours shall be claimed for compensation.

8 Intellectual Property and Ownership

- All source code, documentation, designs, data, and other work products created or populated by the Development Team under this engagement shall be the exclusive property of the Institution. The source code repository must be Institute-controlled at all times. This includes full institutional ownership of the underlying data and codebase, not merely a license to use the Platform.
- The Development Team shall be permitted to reference this project as part of their academic or professional portfolio, subject to prior written consent of DORA regarding the extent of disclosure, and excluding any confidential data.

9 Information Security, Data Privacy, and Confidentiality

- **Information Security & RBAC:** The Platform shall implement strict Role-Based Access Control (RBAC) as defined in Section 2.18. Security measures against unauthorized access, SQL injection, and cross-site scripting must be incorporated.
- **Backup and Disaster Recovery:** Automated daily backups of the database and document repository must be implemented. A disaster recovery plan shall be formulated to restore services with minimal downtime in case of critical failure.
- The Platform will process sensitive data related to institutional, faculty, and student records. Access to production data shall be limited to what is strictly necessary for development, testing, and debugging, and shall follow any data protection guidelines issued by the Institution.
- The Development Team shall not use, disclose, or retain institutional data for any purpose outside the scope of this engagement, both during and after the engagement period.
- **AI Privacy, Grounding, and Auditability:** As the Platform utilizes AI for discrepancy flagging, answers and flags must be grounded in approved institutional data. The model must operate within a strict data boundary, ensuring sensitive personal records are not exposed to external LLMs without contractual safeguards. All AI requests must be logged for auditability, with clear routing for human escalation.

- **Data Minimization and Transparency:** The platform must collect only data required for the approved workflow and maintain a data inventory describing each field, its purpose, retention period, and deletion action.

10 Duration and Termination

- This agreement shall commence on the date of signature by both parties. The Team targets completion and handover of the Platform to DORA by **7 December 2026** (a 14-week period starting 1 September 2026). This target date is subject to change based on more or fewer requirements being added to scope over the course of the engagement (Section 2.20); the Development Team will make its best effort to complete the project as soon as possible.
- In the event of termination, compensation shall be payable for all verified work completed up to the effective date of termination, following Section 7.

11 Post-Deployment Maintenance and Long-Term Sustainability

11.1 Responsibility Allocation Matrix

Once the 14-week development period concludes and the Platform is handed over in production, the ongoing maintenance and governance must be distributed according to the following institutional owners:

Domain	Accountable Institutional Owner
Business Policy & Content	**DORA:** Responsible for deciding NIRF/QS reporting logic, nodal officer rosters, and resolving flagged discrepancies.
Platform Operations	**IIT Mandi IT Services (or designated technical unit):** Responsible for managing server hosting, domain names, periodic data backups, and technical incident response.
Data Protection & Security	**IIT Mandi Security Authority:** Responsible for annual penetration testing, access reviews, and compliance governance.

11.2 Recurring Costs and Sustainability

The estimated budget in Section 7 covers the *one-time development cost*. To ensure the Platform's long-term sustainability (Years 1–3 and beyond), DORA and the Finance office must provision an annual operating budget to cover the following recurring infrastructure costs:

- Managed relational database hosting and AWS S3 (or equivalent) secure object storage.
- Institutional domain names, TLS/SSL certificates, and transactional email/SMS service quotas.
- API token usage costs for the AI-Assisted Discrepancy Flagging models (based on a finalized cost ceiling).

12 Acceptance and Signatures

By signing below, both parties acknowledge that they have read, understood, and agree to the terms described in this document, including the understanding that the technical scope in Section 2 is indicative and will evolve iteratively as described in Section 3.1.

For the Office of the Dean of Resources & Alumni Affairs (DORA), IIT Mandi

_____

Name:_____

Designation:_____

Date:_____

For the Student Development Team

_____

Name:_____
Designation: Team Lead
Date:_____

_____

Name:_____
Designation: Team Lead
Date:_____

Other Team Members (Acknowledgement of Terms):

1. _____ Signature: _____

2. _____ Signature: _____

A Appendix A: Milestone and Deliverable Matrix

Phase	Key Deliverables	Acceptance Criteria	Effort	Timeline
Phase 1: Core NIRF Workflow	End-to-end NIRF data collection, validation, and export platform	System generates a valid NIRF Excel export from mock inputs; maker-checker workflow correctly records audit logs.	80 hrs/week	Weeks 1–4
Phase 2: Analytics	Executive dashboards and QS/THE schema mappings	Dashboards correctly visualize historical data; export works for at least one additional ranking agency.	80 hrs/week	Weeks 5–8
Phase 3: Public & Hardening	Public ranking visualizations, historical query, and reminders	Public portal accessible without login; automated email reminders trigger on deadlines.	60 hrs/week	Weeks 9–11
Phase 4: UAT & Handover	Production deployment, technical/user documentation, and training	DORA signs off on UAT; GitHub repo handed over; staff training sessions completed successfully.	40 hrs/week	Weeks 12–14

Table 1: One-Page Milestone and Deliverable Matrix

B Appendix B: Proposed System Architecture

Frontend (Client Layer):

- Built using Next.js/React.js to provide a highly responsive, dynamic user experience across desktop and mobile form factors.
- Implementation of a strict Role-Based Access Control (RBAC) routing mechanism to securely isolate public views, authenticated Nodal Officer portals, and privileged Admin/Executive dashboards.
- Utilization of institutional navy-and-saffron branding and consistent UX patterns (e.g., empty-state guidance, loading states, and multi-step form confirmations).

Backend (Application Layer):

- Next.js API Routes (Node.js) architected for idempotency, ensuring secure handling of data ingestion, maker-checker validation workflows, and dynamic multi-agency schema mapping logic.
- Resilient document parsing engine built to process multimodal submissions (Excel, PDF, Word) safely in isolated environments to mitigate malware risks.
- Integration with Institute's unified authentication and Single Sign-On (SSO) systems, enforcing Multi-Factor Authentication (MFA) for all privileged administrative accounts.

Data, Storage, and Security Layer:

- A managed Relational Database (e.g., PostgreSQL) designed with a robust schema-migration process. It stores structured ranking metrics, user rosters, and append-only immutable audit logs to guarantee strict data provenance.
- Encrypted Secure Object Storage (e.g., AWS S3 or an Institute-provisioned equivalent) enforcing restricted access URLs with expiring tokens for the centralized document proof repository.
- End-to-end security measures including TLS for data-in-transit, encryption-at-rest for sensitive database fields, and isolated environment management for credentials.
- A defined Disaster Recovery Plan (DRP) including automated, encrypted daily backups designed to meet a target Recovery Point Objective (RPO) of 24 hours and a Recovery Time Objective (RTO) of 8 hours, subject to Institute approval.

Deployment, Hosting, and CI/CD Operations:

- Production deployment on Institute-provisioned servers maintaining absolute data sovereignty. Network access will be restricted to designated campus IPs where mandated.
- A fully Institute-controlled GitHub repository serving as the single source of truth for version control, issue tracking, and architectural documentation.
- An automated CI/CD pipeline bridging separate testing, staging, and production environments. Every pull request will trigger automated dependency/vulnerability scanning (SAST/DAST) before merge.
- Implementation of structured operational runbooks specifying support hours, severity levels, escalation contacts, and predefined release rollback procedures.